Unit: Unit 0: Foundations of Algebra
Topic: Rational Numbers Decimals

Name: ____________

Date: __________

Quiz (Carolina Reaper)

Q1. A cargo ship travels $\frac{3}{4}$ of the distance to its destination in $\frac{2}{3}$ of the total time. What fraction of the total distance does it cover in the remaining $\frac{1}{3}$ of the time?
(A) $\frac{1}{12}$
(B) $\frac{1}{6}$
(C) $\frac{1}{4}$
(D) $\frac{1}{3}$
(E) $\frac{1}{2}$

Q2. A traffic flow model predicts that $\frac{2}{5}$ of the drivers will take a detour, while $\frac{1}{10}$ will take the main road. What percent of drivers will take the main road?
(A) 10
(B) 15
(C) 20
(D) 25
(E) 30

Q3. A delivery truck travels 2.5 miles in $\frac{1}{2}$ hour. What is its average speed in miles per hour?
(A) 3 mph
(B) 4 mph
(C) 5 mph
(D) 6 mph
(E) 8 mph

Q4. A logistics company has $\frac{3}{8}$ of its fleet of trucks currently in use. If $\frac{1}{4}$ of the fleet is being serviced, what fraction of the fleet is available for new shipments?
(A) $\frac{1}{8}$
(B) $\frac{1}{4}$
(C) $\frac{3}{8}$
(D) $\frac{1}{2}$
(E) $\frac{5}{8}$

Q5. A toll road has a $\frac{3}{5}$ chance of being congested during rush hour. What is the probability that it will not be congested?
(A) $\frac{1}{5}$
(B) $\frac{2}{5}$
(C) $\frac{3}{5}$
(D) $\frac{4}{5}$
(E) $\frac{6}{5}$

Q6. An airport has $\frac{2}{3}$ of its flights departing on time. If $\frac{1}{6}$ of the flights are delayed, what fraction of flights are cancelled?
(A) $\frac{1}{6}$
(B) $\frac{1}{4}$
(C) $\frac{1}{3}$
(D) $\frac{2}{3}$
(E) $\frac{3}{4}$

Q7. A transportation company has a $\frac{3}{4}$ full tank of gas. If $\frac{1}{8}$ of the tank is used for a delivery, what fraction of the tank is left?
(A) $\frac{1}{8}$
(B) $\frac{1}{4}$
(C) $\frac{3}{8}$
(D) $\frac{1}{2}$
(E) $\frac{5}{8}$

Q8. A ship travels 2 miles in $\frac{1}{4}$ hour. What is its average speed in miles per hour?
(A) 4 mph
(B) 6 mph
(C) 8 mph
(D) 10 mph
(E) 12 mph

Open Ended Questions (FRQ)

Q9. A delivery company has two routes: Route A is $\frac{3}{5}$ of the total distance and takes $\frac{2}{3}$ of the total time, while Route B is $\frac{2}{5}$ of the total distance and takes $\frac{1}{3}$ of the total time. Compare the average speeds of the two routes and explain your reasoning.

Q10. A logistics company uses a rational number x to represent the ratio of the number of packages delivered to the number of packages shipped. If $x = \frac{a}{b}$, where a and b are integers, and the company delivers $\frac{3}{4}$ of its shipments, what is the value of x? Explain your reasoning and show all work.

Chef's Tips

- Read each question carefully before answering.
- Show all work and explain your reasoning for free response questions.
- Use a calculator only when necessary and be sure to show all work.